

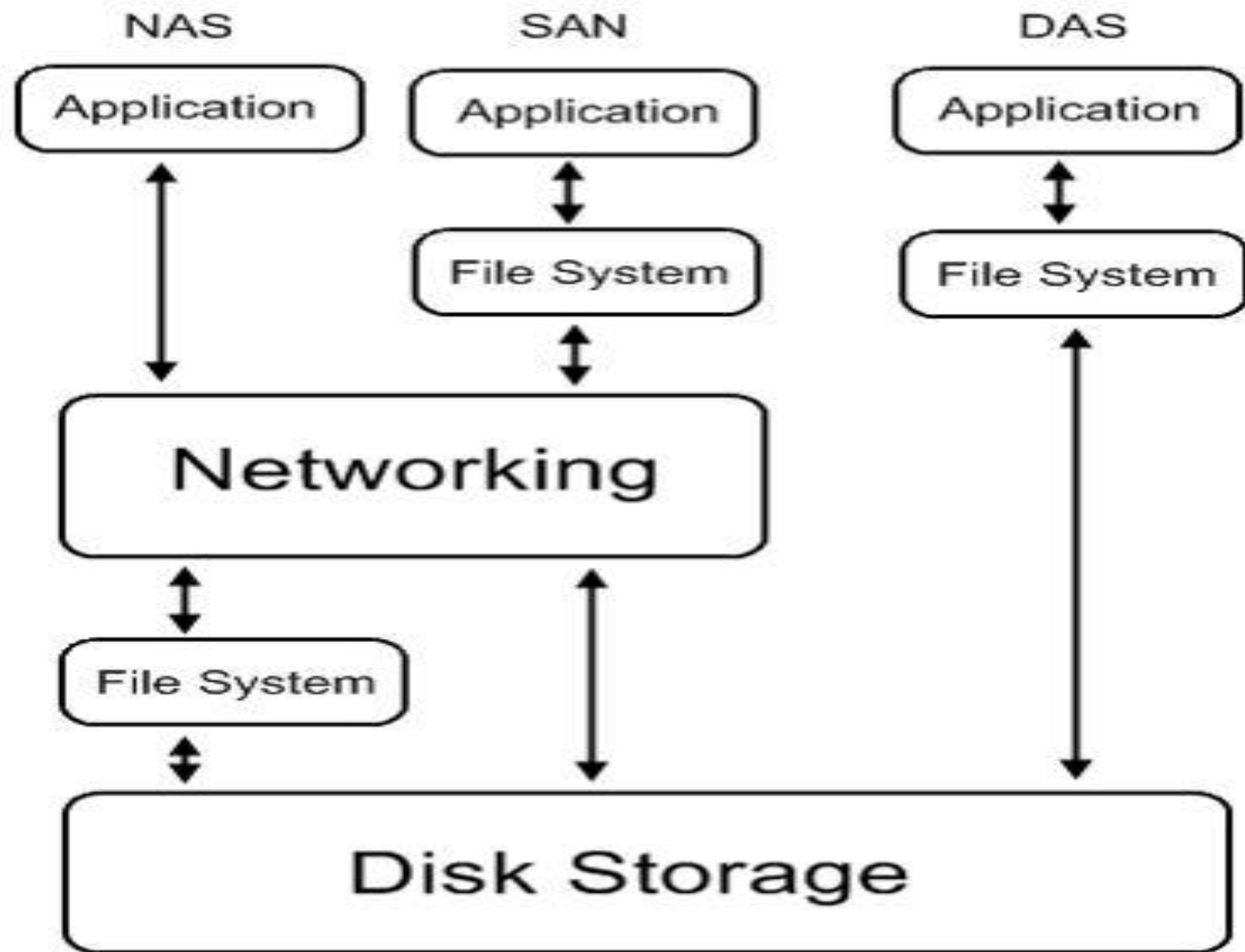


MINIA UNIVERSITY

Storage Environment

Storage Environment

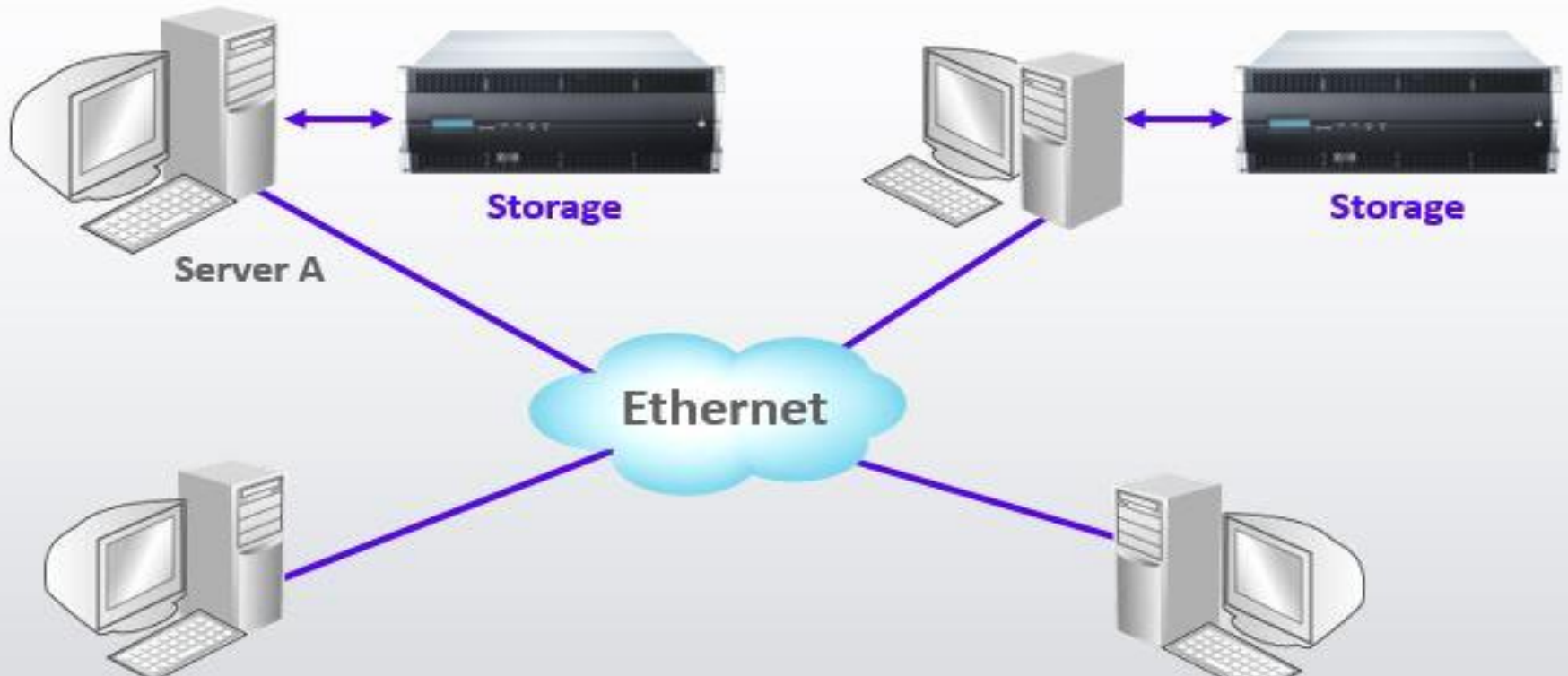
- **DAS**
Direct Attached Storage
- **SAN**
Storage Area Networks
- **NAS**
Network Attached Storage



DAS - Direct Attached Storage

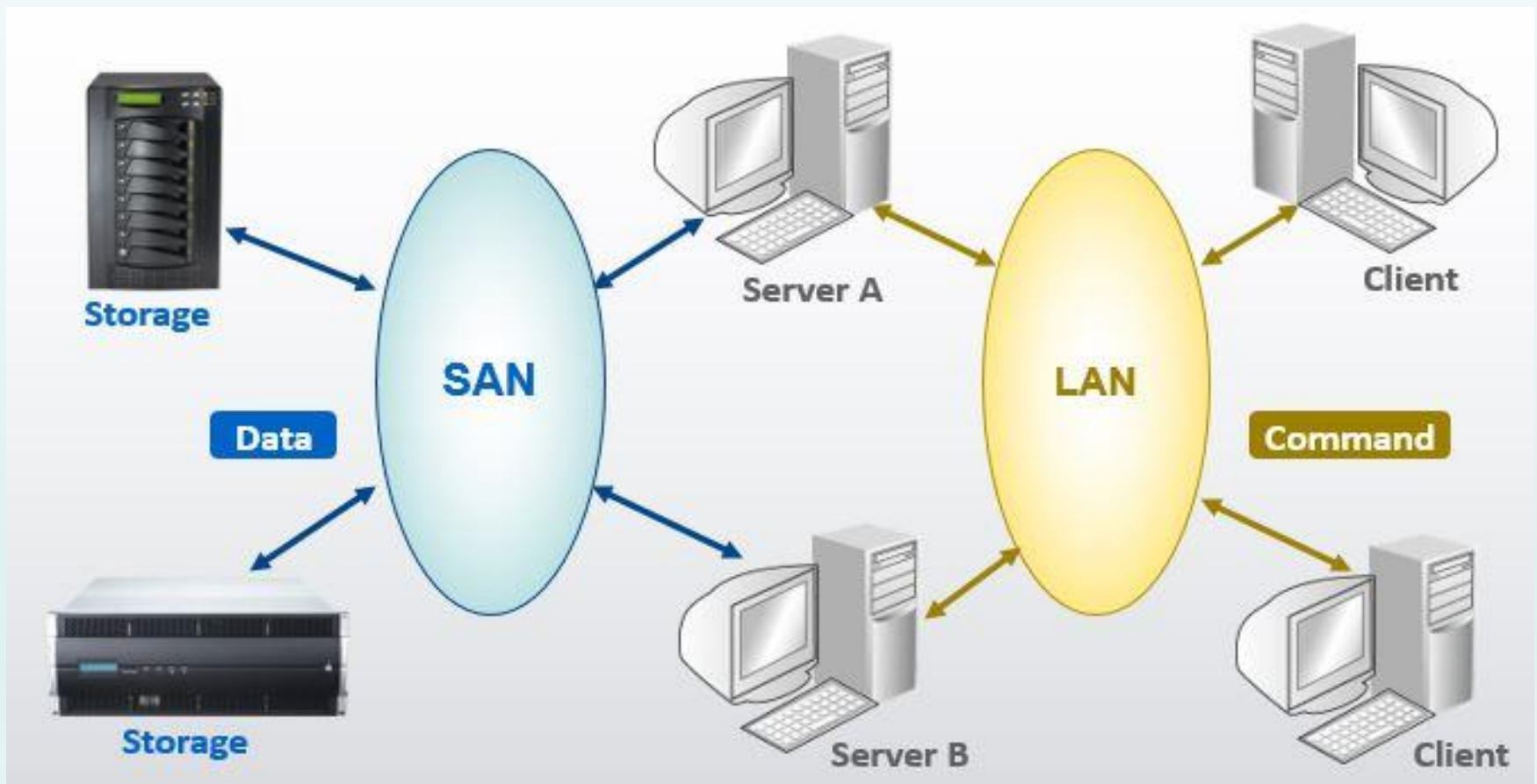
Data storage connect **directly** to a server through **Host Bus Adapter (HBA)**, there is **no network** between storage and host servers

- SCSI - RAID



SAN - Storage Area Networks

- A dedicated data storage network which can be accessed by **multiple servers**



SAN - Storage Area Networks

- A SAN is a way to provide users shared access to consolidated.
- **Block level** data storage,
- even allowing multiple clients to access files at the same time with **very high performance**.
- Protocols that are used in SAN are:
SCSI, SATA

SAN - Storage Area Networks

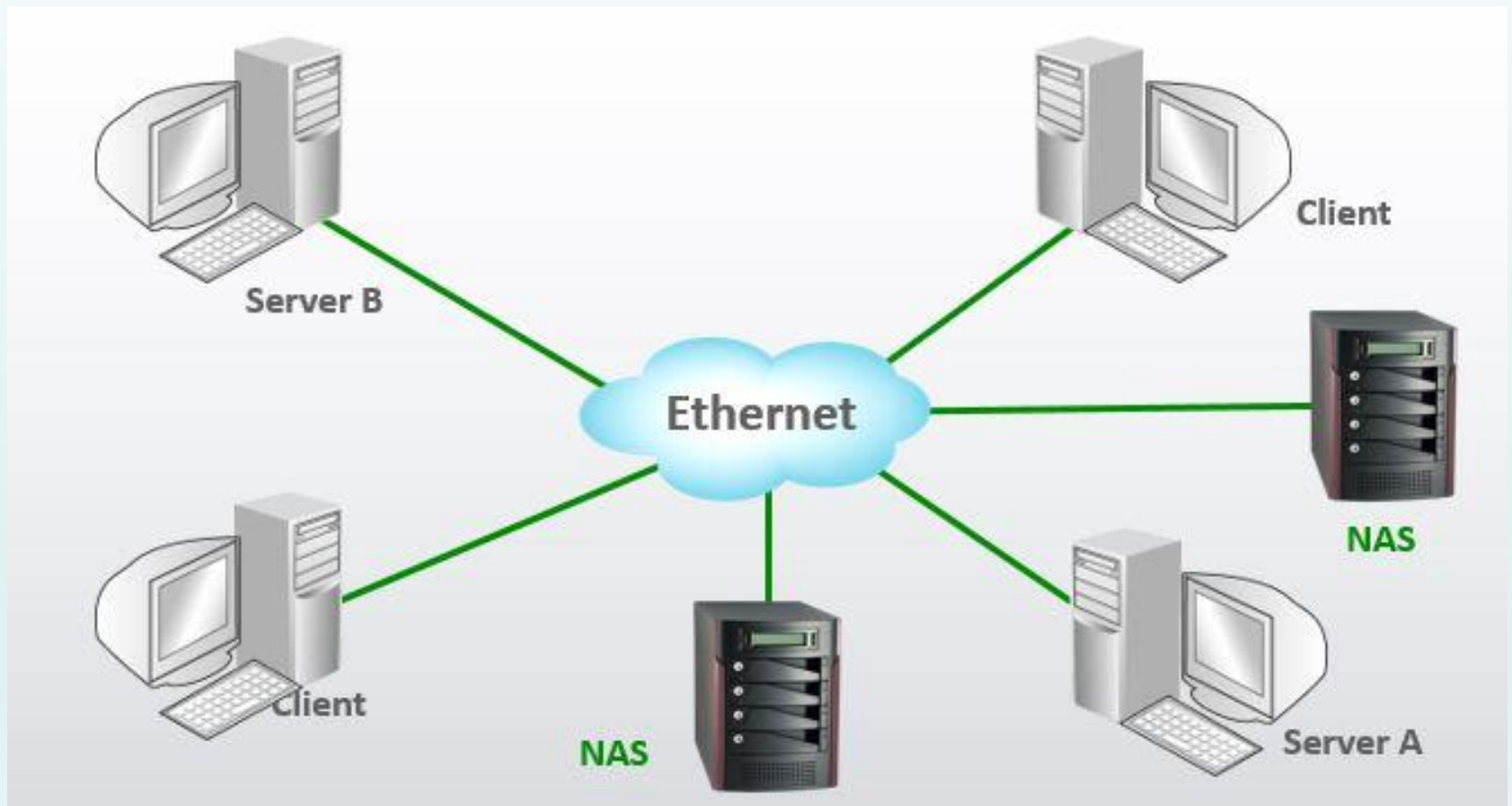
- Advantage: no distance limitation
- Disadvantage: Speed and Latency
- Weakness: Security

SAN - Component

- Node ports
- Cables
- **Interconnect device** such as: Hubs, switches, directors
- Storage arrays
- SAN management Software

NAS - Network Attached Storage

- A file level computer data storage connect to other device on TCP/ IP network



NAS - Network Attached Storage Component

- **1.** Head unit: CPU, Memory
- **2.** Network Interface Card (NIC)
- **3.** Optimized operating system
- **4.** Protocols
- **5.** Storage protocols: ATA, SCSI, FC

NAS - Benefits

- Relatively inexpensive
- 24/7 and remote data availability
- Good expandability
- Redundant storage architecture
- Automatic backups to other devices and cloud
- Flexibility

	DAS	NAS	SAN
Storage Type	sectors	shared files	blocks
Data Transmission	IDE/SCSI	TCP/IP, Ethernet	<u>Fibre Channel</u>
Access Mode	clients or servers	clients or servers	servers
Capacity (bytes)	10^9	$10^9 - 10^{12}$	$\gg 10^{12}$
Complexity	Easy	Moderate	Difficult
Management Cost (per GB)	High	Moderate	Low

Redundant Array of Independent Disks (RAID)

- A group of hard disks is called a **disk array**
- RAID combines a disk array into a single virtual device
 - called RAID drive
- Provide fault tolerance for shared data and applications
- Different implementations: **Level 0-5**
- Characteristics:
 - **Storage** Capacity
 - **Speed**: Fast Read and/or Fast Write
 - **flexibility** in the face of device failure

Start here

RAID Functions

- **Striping**

- Write consecutive logical byte/blocks on consecutive physical disks

يقطعها في شكل بلوكات أو بايتس .

- **Mirroring**

- Write the same block on two or more physical disks

بيّنسخ نفس البلك

- **Parity Calculation**

بيّنسخ الملف كله مرة واحدة

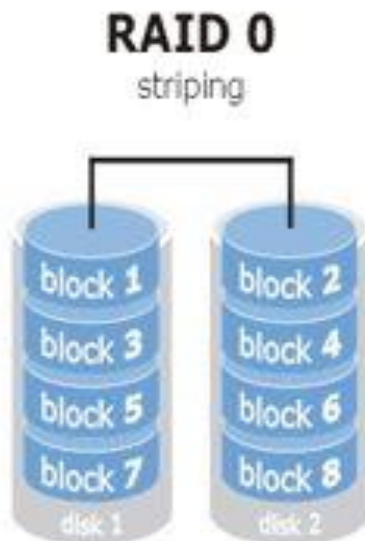
RAID Types

- **RAID 0**
 - Stripe with no parity (see next slide for figure)
- **RAID 1**
 - Mirror two or more disks
- **RAID 0+1**
 - Stripe and Mirrors
- **RAID 3**
 - Synchronous, Subdivided Block Access; Dedicated Parity Drive
- **RAID 5**
 - Like RAID 4, but parity striped across multiple drives

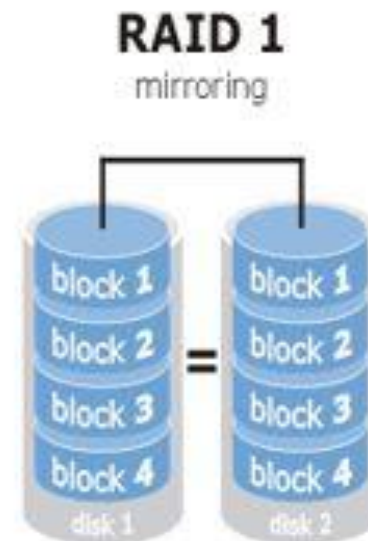
RAID 10

RAID Types

RAID 0



RAID 1



RAID 0 – Blocks Striped. No Mirror. No Parity.

Disk Striping (no redundancy)

RAID 1 – Blocks Mirrored. No Stripe. No parity.

RAID 0

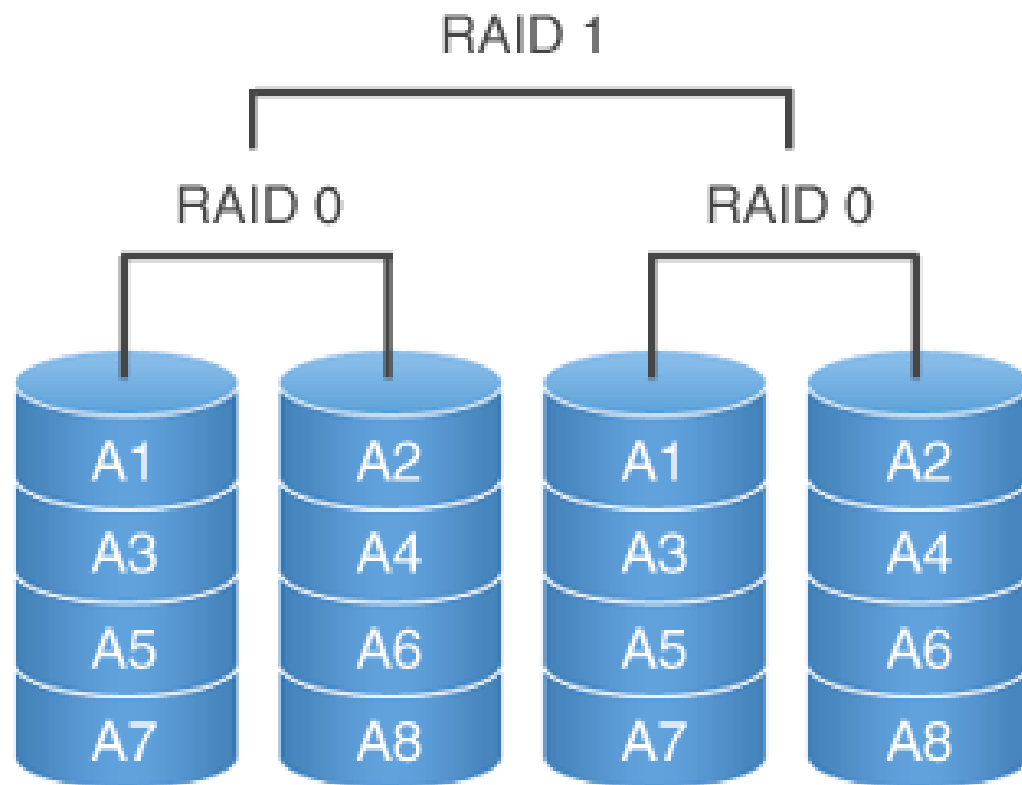
- Minimum 2 disks.
- Excellent performance (as blocks are striped).
- No redundancy (no mirror, no parity).
- Don't use this for any critical system.

stripe is related to performance
mirror related to redundancy

RAID 1

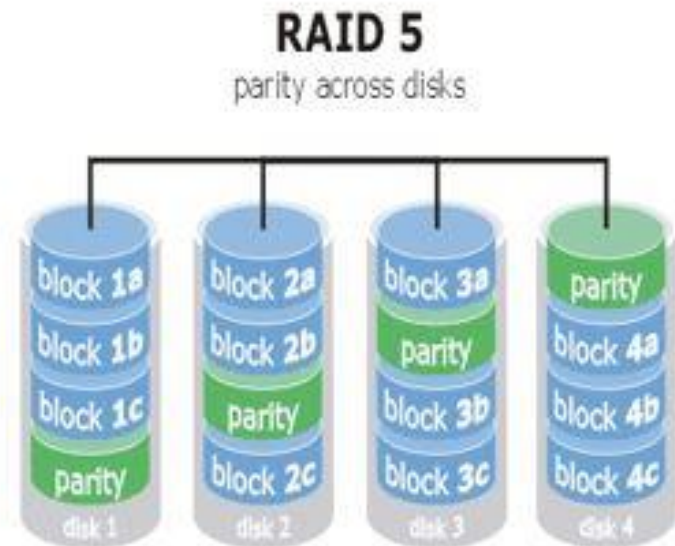
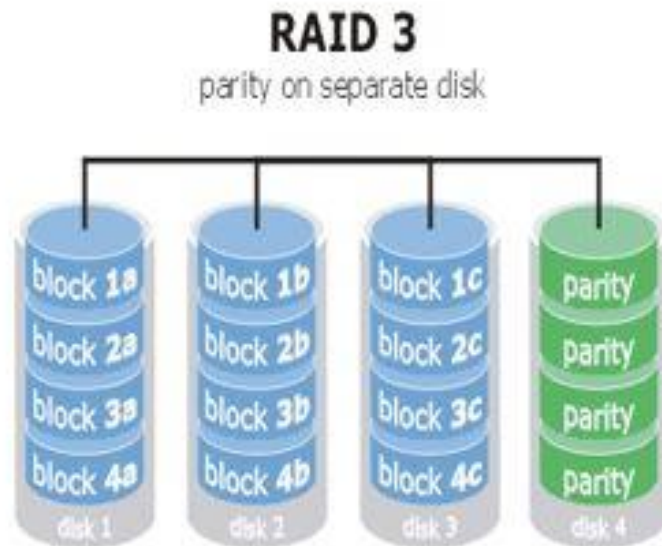
- Minimum 2 disks.
- Good performance (no striping. no parity).
- Excellent redundancy (as blocks are mirrored).

RAID 0+1



RAID 3

RAID 5



RAID 3 – Bytes Striped. (and Dedicated Parity Disk)

Disk striping with Dedicated Parity Drive

Disk striping with Distributed Parity Data

RAID 5 – Blocks Striped. Distributed Parity.

RAID 3, RAID4

- **RAID 3** : This uses byte level striping
- **RAID 4** : uses block level striping

Minimum of 3 disks (2 disks for data and 1 for parity)

RAID 4 : is not commonly used.

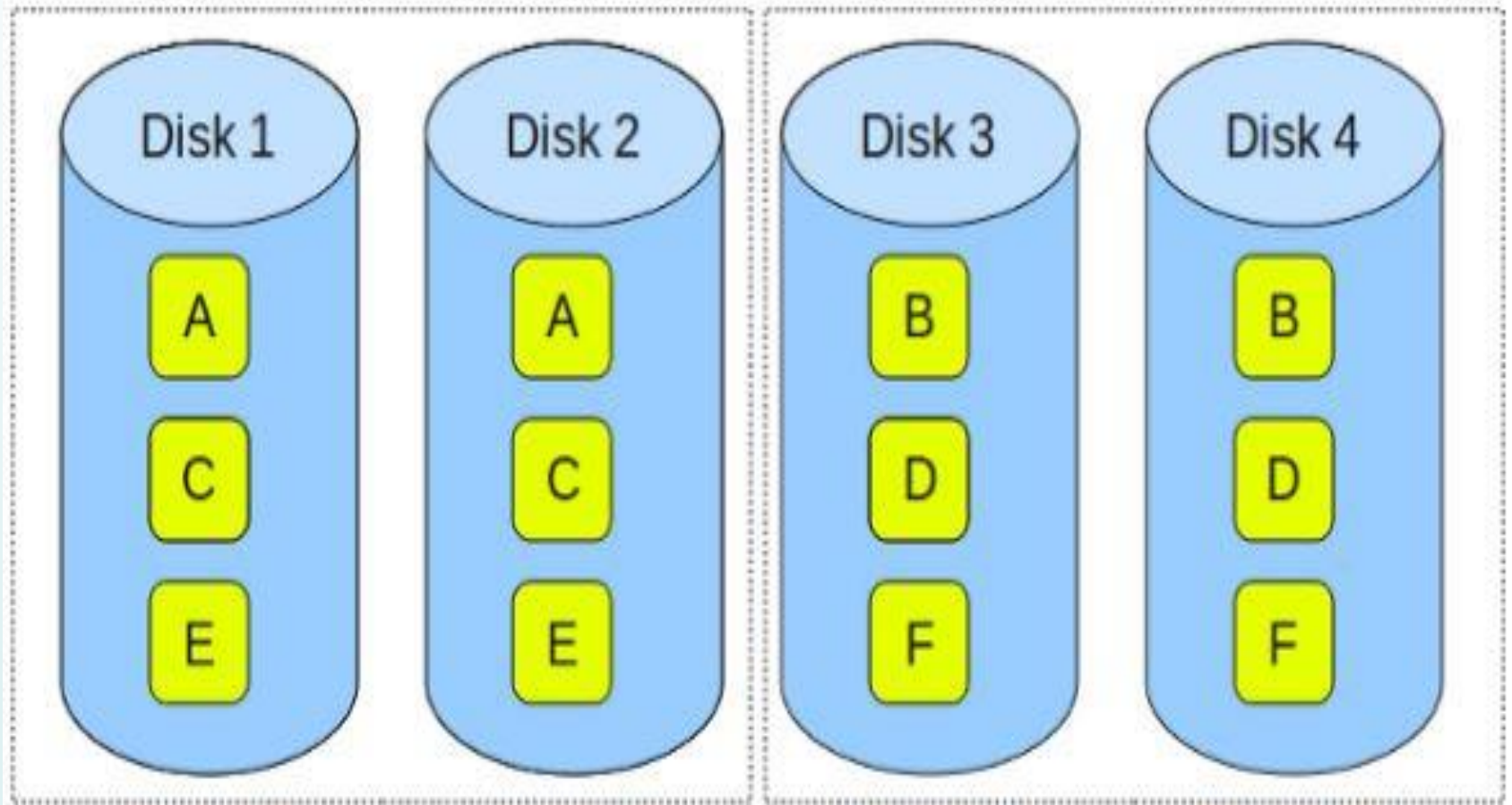
RAID 5

- Minimum 3 disks.
- Good performance (as blocks are striped).
- Good redundancy (distributed parity).
- Best cost effective option providing both performance and redundancy. Use this for DB that is heavily read oriented. Write operations will be slow.

RAID 6

- Similar RAID 5 except:
- It has double parity blocks
- Require 6 disk

RAID 10



RAID 10 – Blocks Mirrored. (and Blocks Striped)

RAID 10

- Minimum 4 disks.
- This is also called as “stripe of mirrors”
- Excellent redundancy (as blocks are mirrored)
- Excellent performance (as blocks are striped)
- If you can afford the dollar, this is the BEST option for any mission critical applications (especially databases).

RAID 0 splits data across drives, resulting in higher data throughput. The performance of this configuration is extremely high, but a loss of any drive in the array will result in data loss. This level is commonly referred to as striping.

Minimum number of drives required: 2

Performance: High

Redundancy: Low

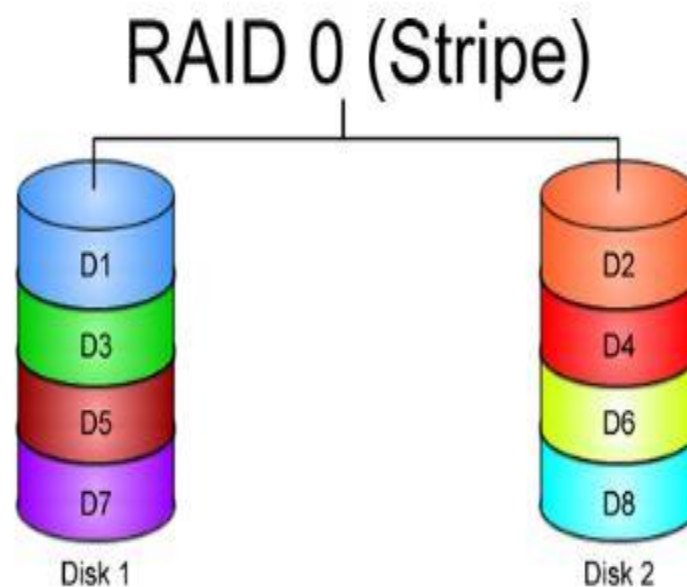
Efficiency: High

ADVANTAGES:

- High performance
- Easy to implement
- Highly efficient (no parity overhead)

DISADVANTAGES:

- No redundancy
- Limited business use cases due to no fault tolerance



RAID 1 writes all data to two or more drives for 100% redundancy: if either drive fails, no data is lost. Compared to a single drive, RAID 1 tends to be faster on reads, slower on writes. This is a good entry-level redundant configuration. However, since an entire drive is a duplicate, the cost per megabyte is high. This is commonly referred to as mirroring.

Minimum number of drives required: 2

Performance: Average

Redundancy: High

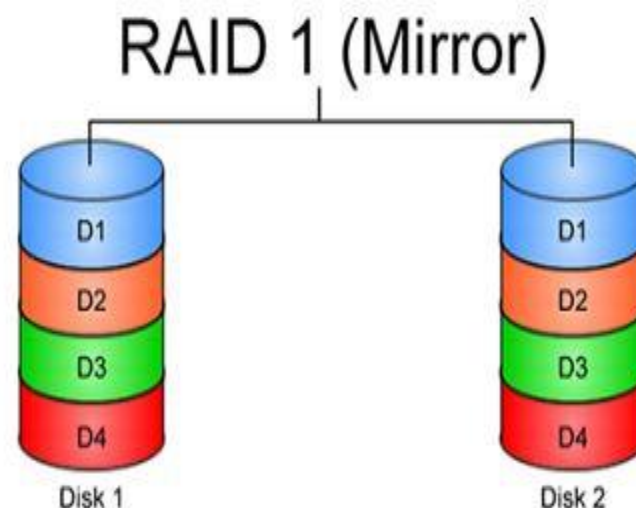
Efficiency: Low

ADVANTAGES:

- Fault tolerant
- Easy to recover data in case of drive failure
- Easy to implement

DISADVANTAGES:

- Highly inefficient (100% parity overhead)
- Not scalable (becomes very costly as number of disks increase)



Minimum number of drives required: 4

Performance: Very High

Redundancy: High

Efficiency: Low

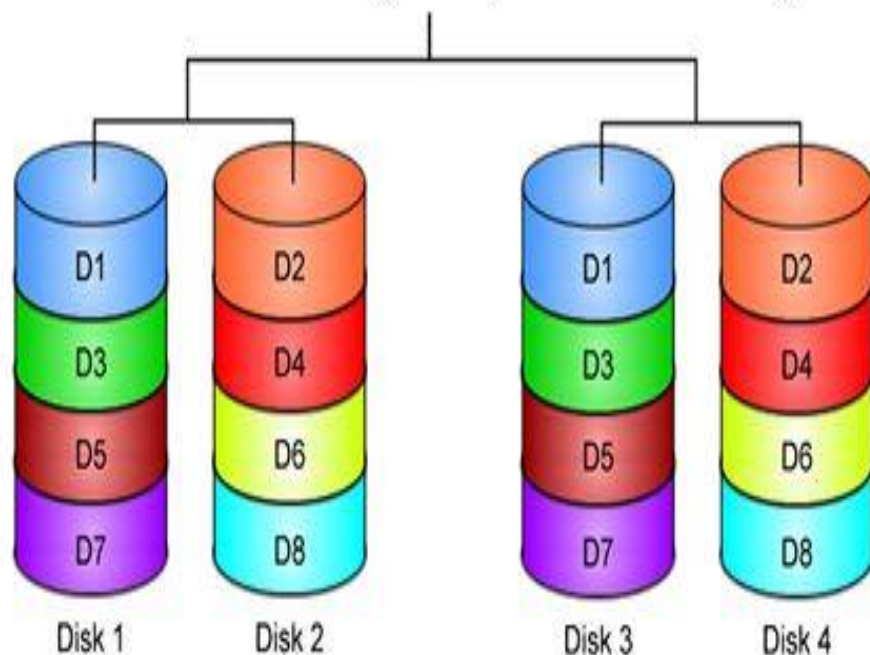
ADVANTAGES:

- Fault tolerant
- Very high performance

DISADVANTAGES:

- Expensive
- High Overhead
- Very limited scalability

RAID 0+1 (Stripe+Mirror)



RAID 5 stripes data at a block level across several drives, with parity equally distributed among the drives. The parity information allows recovery from the failure of any single drive. Write performance is rather quick, but because parity data must be skipped on each drive during reads, reads are slower. The low ratio of parity to data means low redundancy overhead.

Minimum number of drives required: 3

Performance: Average

Redundancy: High

Efficiency: High

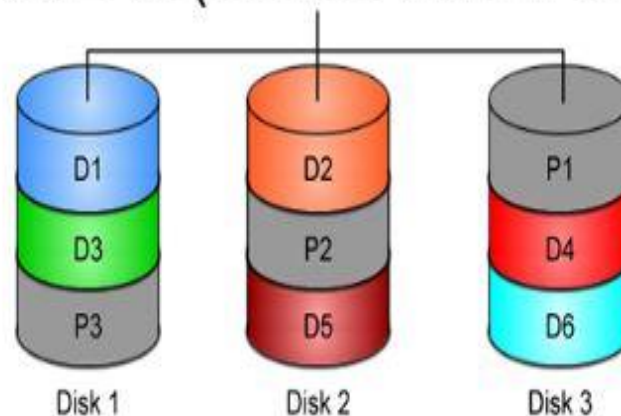
ADVANTAGES:

- Fault tolerant
- High efficiency
- Best choice in multi-user environments which are not write performance sensitive

DISADVANTAGES:

- Disk failure has a medium impact on throughput
- Complex controller design

RAID 5 (Drives with Parity)



RAID 10 is a striped (RAID 0) array whose segments are mirrored (RAID 1). RAID 10 is a popular configuration for environments where high performance and security are required. In terms of performance it is similar to RAID 0+1. However, it has superior fault tolerance and rebuild performance.

Minimum number of drives required: 4

Performance: Very High

Redundancy: Very High

Efficiency: Low

ADVANTAGES:

- Extremely high fault tolerance
 - under certain circumstances, RAID 10 array can sustain multiple simultaneous drive failures
- Very high performance
- Faster rebuild performance than 0+1

DISADVANTAGES:

- Very Expensive
- High Overhead
- Limited scalability

RAID 10 (Mirror+Stripe)

